

Learn to Analyse Logs with Elasticsearch, Logstash and Kibana

Elasticsearch, Logstash and Kibana together form the ELK Stack. Elasticsearch is a search and analysis engine. Logstash is a data collecting engine with capabilities of real-time pipelining, and Kibana is a user interface for visualising Elasticsearch data by creating graphs and charts. This article explains the process of analysing the logs generated by IT usage in a company.



Most IT resources generate streams of chronological messages called logs or audit trails that have information related to the systems' activities. Logs are the most important assets for monitoring and managing various IT resources. They provide the information generated by the system, which is required to spot issues that might affect its performance. These logs record every action that has taken place

on your IT resources (here, IT resources means computers, networks and any other IT systems). In every organisation, log analysis should be a part of infrastructure monitoring as it helps mitigate several risks at an early stage, and also helps meet compliance regulations.

How does log analysis work?

To analyse the logs from different IT resources such as network devices and programmable or smart devices, the

following way is recommended.

1. **Collect logs:** First, collect logs from across your infrastructure in a central and accessible location, which you can do with a logging application called Log Collector. It's easier to query and analyse logs from a single location.
2. **Indexing:** When collected at a central location, logs should be organised in a common format to ensure uniformity, to make them more understandable. They should also be indexed by this central location to make the data easily searchable for efficient log analysis.
3. **Search and analyse:** Logs can be searched by matching various patterns and structures such as those with a specific severity level or by exception. Using this search, reports and dashboards can be created to make information available to stakeholders. Of course, it is easier to spot trends by looking at graphs or other visual representations of data that you can view on your reports. These reports and dashboards make log analysis way more easy than going through the log files.
4. **Monitor and set alerts:** Alerts can be set up to notify you when any conditions are not met. This helps to avoid risks by detecting the root cause of the issues that impact the performance of your IT resources,